

2026-05-21 Notes: `v3tk_VRI_image.sh`

Pipeline

Purpose

`v3tk_VRI_image.sh` is the top-level shell wrapper for making the final VRI visual products in this directory.

It starts from existing VRI FITS products:

```
*_DATACUBE_FINAL_WCS_Pall_mad_red_v3tk_VRI.fits
```

and produces, for each galaxy:

```
<GALAXY>_observed_VRI.png  
<GALAXY>_observed_VRI.pdf  
<GALAXY>_legacy_reprojected.jpg  
<GALAXY>_combined_VRI.png
```

The shell script does not create the VRI FITS files. Those are upstream prerequisites produced by `v3tk_to_VRI.py`, normally via `v3tk_to_VRI.sh`.

Main Command

Run the full VRI image pipeline from:

```
cd /Users/Igniz/Desktop/ICRAR/v3tk_to_VRI  
./v3tk_VRI_image.sh
```

The script writes a timestamped log:

```
v3tk_VRI_image_YYYYMMDD_HHMMSS.log
```

Example:

```
v3tk_VRI_image_20260521_131131.log
```

Pipeline Summary

`v3tk_VRI_image.sh` runs three Python scripts:

```
Step 1: v3tk_observed_VRI_image.py
Step 2: v3tk_get_legacy.py
Step 3: v3tk_combined_VRI_image.py
```

Data flow:

```
*_v3tk_VRI.fits
-> *_observed_VRI.png / *_observed_VRI.pdf
-> *_legacy_reprojected.jpg
-> *_combined_VRI.png
```

Then `auto_arrange_and_combine.py` can be run separately to arrange the per-galaxy `*_combined_VRI.png` panels into a single mosaic.

Top-Level Shell Script

File:

```
v3tk_VRI_image.sh
```

Important behavior:

- `set -euo pipefail` makes the script stop on unhandled errors, unset variables, or failed pipe components.
- It changes directory to the directory containing the script, so relative paths are stable.
- It creates a timestamped log file and tees all output to that log.
- It detects CPU count using Python, `getconf`, `nproc`, or `sysctl`.
- It tries to activate the `ICRAR` conda environment.
- If `ICRAR` is not active, it falls back to `conda run -n ICRAR python`.
- It runs all three stages with `--overwrite`.

The script-level controls are:

```
MAX_DL=1 ./v3tk_VRI_image.sh
```

`MAX_DL` limits concurrent Legacy Survey downloads. The default is:

```
MAX_DL=2
```

This throttle matters because the Legacy Survey service can rate-limit simultaneous HTTP requests.

Note: the current shell script prints:

```
Step 1/3: observed R images
```

but the actual command is:

```
v3tk_observed_VRI_image.py
```

So the message is just a stale label; the stage is VRI, not R-only.

Upstream Prerequisite: `v3tk_to_VRI.sh` and `v3tk_to_VRI.py`

Although `v3tk_VRI_image.sh` does not call the converter, its input files come from:

```
v3tk_to_VRI.sh  
v3tk_to_VRI.py
```

`v3tk_to_VRI.sh`:

- logs to `v3tk_to_VRI.log`;
- reads compressed MUSE cube files from:

```
/arc/projects/mauve/cubes/v3tk
```

- copies each `*_v3tk.fits.gz` into the working directory;
- unzips it;

- runs:

```
python v3tk_to_VRI.py <cube.fits>
```

- removes the local uncompressed input cube after conversion.

`v3tk_to_VRI.py` collapses a 3D MUSE cube into V/R/I flux and AB magnitude maps. It writes a VRI FITS file with these HDUs:

```
V_FLUX  
V_MAG  
R_FLUX  
R_MAG  
I_FLUX  
I_MAG
```

Important converter defaults:

- input cube HDU: `DATA`;
- V filter: `bessell-V`;
- R filter: `bessell-R`;
- I filter: `bessell-I`;
- raw cube flux scale: `1e-20 erg/s/cm^2/Angstrom`;
- default NaN policy: replace non-finite cube samples with zero for visualization;
- partial filter overlap is enabled by default, so filters can be truncated and renormalized to the cube wavelength range.

The converter preserves the spatial WCS in the 2D output HDUs.

Step 1: Observed VRI Rendering

File:

```
v3tk_observed_VRI_image.py
```

Command used by the shell script:

```
python v3tk_observed_VRI_image.py --input-dir . --overwrite --workers "$NCPU"
```

Default input pattern:

```
*_DATACUBE_FINAL_WCS_Pall_mad_red_v3tk_VRI.fits
```

For each matched FITS file, it writes:

```
<GALAXY>_observed_VRI.png  
<GALAXY>_observed_VRI.pdf
```

The galaxy ID is parsed from the FITS filename by removing:

```
_DATACUBE_FINAL_WCS_Pall_mad_red_v3tk_VRI.fits
```

Rendering details:

- reads `V_FLUX`, `R_FLUX`, and `I_FLUX`;
- prepares non-finite values as zero;
- clips negative values to zero for display;
- maps channels as:

```
I -> red  
R -> green  
V -> blue
```

- applies a gray-world white balance;
- scales channels using luminance percentiles;
- applies optional gamma;
- uses Astropy Lupton/asinh RGB rendering;
- applies a final post-render brightness boost;
- writes PNG at native image dimensions;
- writes a PDF with the same visual appearance.

Important defaults:

```
percentile-low: 1.0
percentile-high: 99.9
stretch: 1.0
Q: 8.0
gamma: 1.0
post-boost: 2.5
workers: os.cpu_count()
overwrite: true
```

Useful dry run:

```
/opt/miniconda3/envs/ICRAR/bin/python v3tk_observed_VRI_image.py --
dry-run
```

Potential warning:

If Matplotlib cannot write to `~/.matplotlib`, it creates a temporary cache directory. This is not a pipeline failure, but setting `MPLCONFIGDIR` to a writable directory can reduce startup overhead in multiprocessing runs.

Example:

```
MPLCONFIGDIR=/private/tmp/mplconfig ./v3tk_VRI_image.sh
```

Step 2: Legacy Survey Download and Reprojection

File:

```
v3tk_get_legacy.py
```

Command used by the shell script:

```
python v3tk_get_legacy.py --overwrite --workers "$NCPU" --max-
concurrent-downloads "$MAX_DL"
```

For each matched VRI FITS file, it writes:

```
<GALAXY>_legacy_reprojected.jpg
```

Temporary cutout files:

```
<GALAXY>_Legacy.jpg  
<GALAXY>_Legacy.fits
```

By default, the temporary JPEG is deleted after successful reprojection, and the temporary FITS is deleted immediately after WCS use. Use `--keep-cutout` if the downloaded JPEG should be retained.

What the script does:

- selects a 2D flux HDU from the VRI FITS, preferring `R_FLUX`;
- gets the target MUSE WCS and shape;
- checks that `<GALAXY>_observed_VRI.png` matches the FITS image size;
- computes the image-center sky coordinate;
- estimates the MUSE footprint size from the WCS corners;
- requests a Legacy Survey cutout centered on the MUSE field;
- downloads both a JPEG cutout and a FITS cutout;
- uses the FITS cutout header for authoritative Legacy WCS;
- verifies that the cutout covers the target MUSE footprint;
- enlarges the requested cutout up to three attempts if coverage is insufficient;
- reprojects each RGB channel to the MUSE grid using `reproject_adaptive(conservative=True)`;
- writes the reprojected RGB image as JPEG.

Important defaults:

```
layer: ls-dr9  
pixscale: 0.262 arcsec/pixel  
margin-factor: 1.02  
timeout: 120 seconds  
workers: min(2, os.cpu_count()) when run directly  
max-concurrent-downloads: 2  
overwrite: true
```

The shell script passes `--workers "$NCPU"`, but download concurrency is separately limited by `MAX_DL`.

Useful dry run:

```
/opt/miniconda3/envs/ICRAR/bin/python v3tk_get_legacy.py --dry-run
```

If HTTP 429 or 503 occurs, the downloader retries with exponential backoff and random jitter. If rate limits continue, run:

```
MAX_DL=1 ./v3tk_VRI_image.sh
```

Step 3: Combine Observed MUSE and Legacy Background

File:

```
v3tk_combined_VRI_image.py
```

Command used by the shell script:

```
python v3tk_combined_VRI_image.py --overwrite --workers "$NCPU"
```

For each matched VRI FITS file, it requires:

```
<GALAXY>_observed_VRI.png  
<GALAXY>_legacy_reprojected.jpg
```

and writes:

```
<GALAXY>_combined_VRI.png
```

Combination logic:

- reads a 2D flux HDU from the VRI FITS, preferring `R_FLUX`;
- builds a valid-pixel mask:

```
finite and > 0
```

- flips the mask vertically to match the PNG orientation written by `v3tk_observed_VRI_image.py`;

- starts from the Legacy reprojected image;
- replaces valid mask pixels with observed VRI RGB pixels;
- writes the result as PNG.

The combined image therefore shows:

- MUSE observed VRI rendering where the VRI flux map is valid;
- Legacy Survey background where the MUSE rendering is invalid, blank, NaN, or non-positive.

Size checks:

- observed PNG must match the VRI FITS 2D shape;
- legacy reprojected JPG must match the VRI FITS 2D shape;
- mismatch raises an error rather than silently combining misaligned images.

Useful dry run:

```
/opt/miniconda3/envs/ICRAR/bin/python v3tk_combined_VRI_image.py --  
dry-run
```

Current Workspace Snapshot

On 2026-05-21, this directory contains 26 VRI FITS inputs matching:

```
*_DATACUBE_FINAL_WCS_Pall_mad_red_v3tk_VRI.fits
```

The same directory also contains existing per-galaxy products such as:

```
IC3392_observed_VRI.png  
IC3392_observed_VRI.pdf  
IC3392_legacy_reprojected.jpg  
IC3392_combined_VRI.png
```

and equivalent products for other galaxies.

Relevant Files

Top-level runner:

```
v3tk_VRI_image.sh
```

Upstream VRI FITS generation:

```
v3tk_to_VRI.sh  
v3tk_to_VRI.py
```

Pipeline stages called by `v3tk_VRI_image.sh`:

```
v3tk_observed_VRI_image.py  
v3tk_get_legacy.py  
v3tk_combined_VRI_image.py
```

Optional mosaic arranger after the per-galaxy combined images are made:

```
auto_arrange_and_combine.py
```

Related R-band analogues in the repository:

```
v3tk_R_image.sh  
v3tk_observed_R_image.py  
v3tk_combined_R_image.py
```

These are not run by `v3tk_VRI_image.sh`, but they follow a similar pattern for R-only products.

Dependency State

The validated Python environment is:

```
/opt/miniconda3/envs/ICRAR/bin/python
```

Relevant package versions checked on 2026-05-21:

```
numpy 2.2.6
astropy 7.0.2
speclite 0.20
matplotlib 3.10.3
Pillow 11.3.0
reproject 0.14.1
```

Stage-specific dependencies:

- `v3tk_to_VRI.py`: `numpy`, `astropy`, `speclite`;
- `v3tk_observed_VRI_image.py`: `numpy`, `astropy`, `matplotlib`, `Pillow`;
- `v3tk_get_legacy.py`: `numpy`, `astropy`, `reproject`, `Pillow`, Python `urllib`;
- `v3tk_combined_VRI_image.py`: `numpy`, `astropy`, `Pillow`.

Failure Modes and Checks

No VRI FITS matched:

```
No files matched pattern
'*_DATACUBE_FINAL_WCS_Pall_mad_red_v3tk_VRI.fits'
```

This means the upstream conversion has not been run, the script is in the wrong directory, or the pattern differs.

Missing observed PNG during Legacy step:

```
Regenerate observed PNGs from the current FITS with: python
v3tk_observed_VRI_image.py
```

This can happen if Step 2 is run independently before Step 1.

Size mismatch:

The scripts intentionally fail if FITS dimensions, observed PNG size, and reprojected Legacy image size do not agree. This protects against combining products from different runs or stale files.

Download rate limiting:

The Legacy Survey server may return rate-limit or temporary service errors. The downloader retries, but `MAX_DL=1` is safer if repeated HTTP errors occur.

Network required:

Step 2 requires internet access to:

```
https://www.legacysurvey.org/viewer/jpeg-cutout  
https://www.legacysurvey.org/viewer/fits-cutout
```

Recommended Routine

If VRI FITS products already exist:

```
cd /Users/Igniz/Desktop/ICRAR/v3tk_to_VRI  
MAX_DL=2 ./v3tk_VRI_image.sh
```

If Legacy Survey rate-limits the run:

```
MAX_DL=1 ./v3tk_VRI_image.sh
```

After per-galaxy combined images exist, make a combined mosaic:

```
/opt/miniconda3/envs/ICRAR/bin/python auto_arrange_and_combine.py  
*combined_VRI.png 16 9
```

For a quick mosaic without proof:

```
/opt/miniconda3/envs/ICRAR/bin/python auto_arrange_and_combine.py -  
-fast *combined_VRI.png 16 9
```

Important Scientific and Display Assumptions

The VRI FITS files are visualization products, not calibrated standard broadband photometry in every case. Partial filter overlap is enabled by default in `v3tk_to_VRI.py`, so if the MUSE cube does not fully cover a Bessell filter, the filter response is truncated and renormalized to the cube wavelength range.

The observed PNG/PDF products are display renderings:

- non-finite samples are handled for visualization;
- negative display values are clipped;
- gray-world white balance and Lupton/asinh stretch affect visual color;
- `post_boost=2.5` brightens the final RGB image.

The combined image is a visual overlay product:

- MUSE VRI rendering is used where `R_FLUX` is finite and positive;
- Legacy Survey is used elsewhere;
- the mask is based on the selected flux map, not on all three V/R/I channels simultaneously.

These choices are appropriate for visually inspecting the MUSE footprint against a Legacy Survey background, but they should not be interpreted as a direct calibrated RGB photometric product without checking the underlying FITS maps.